The multiple pendulum as a beaded string: hidden tridiagonal structure from normal modes to nonlinear dynamics

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Abstract

The equations of motion of a planar pendulum made of n point masses joined by massless rods look forbidding in the usual angle coordinates: through a dense mass matrix, every mass is coupled to every other. We show that this coupling is an artifact of the coordinates. A one-line factorization of the mass matrix reveals that, in terms of the horizontal displacements of the masses, the small oscillations are those of a beaded string whose tension decreases with height, and that in the angle coordinates themselves the inverse mass matrix and the dynamical matrix are tridiagonal, with simple closed-form entries. Classical results on tridiagonal (Jacobi) matrices then give, with almost no work, simple frequencies and modes whose k-th member has $k-1$ nodes, and the spectrum converges to that of the continuous hanging chain; numerically, the discrete chain behaves like a continuous one lengthened by half a link. For large amplitudes, the rod tensions obey a tridiagonal linear system whose zero-angle limit is the inverse mass matrix of the linear theory rescaled by the rod lengths, so the exact nonlinear accelerations follow from a few lines of code at a cost proportional to n. The material connects introductory mechanics, linear algebra and computational physics and lends itself to classroom and computer exercises.

1 Introduction

The double pendulum is the standard first example of chaos in mechanics, and chains of pendulums are used to model ropes, chains and even ponytails.[1–3] Students who try to go beyond two or three links, however, quickly meet equations of motion that seem to couple every mass to every other. In the absolute angles θ_k of the links, the mass matrix of an n-mass pendulum is dense: each entry is a sum of masses times a cosine of an angle difference.[4–7]

Physically this is puzzling. Each mass is acted on only by gravity and by the two rods attached to it, so the dynamics should couple each link only to its neighbors. This paper explains where the nearest-neighbor structure is hidden and what follows from it. The central observation is a factorization of the mass matrix (Sec. 2) that exposes, after a change of variables, a familiar system: a beaded string under nonuniform tension (Sec. 3). Back in angle coordinates, the inverse mass

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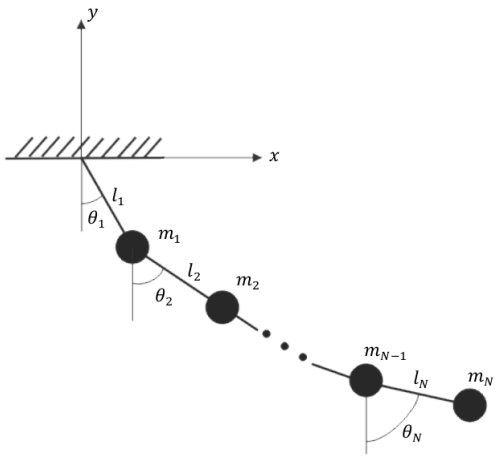


Figure 1: Planar pendulum with n point masses m_k and massless rods of lengths l_k. Each angle θ_k is measured from the downward vertical.

matrix and the dynamical matrix are tridiagonal in closed form. The normal modes then follow from the classical theory of tridiagonal matrices, and the continuum limit reproduces the hanging chain (Sec. 4). For large amplitudes, a Newtonian formulation with the rod tensions as unknowns again leads to a tridiagonal system, and its zero-angle limit is the inverse mass matrix of the linear theory rescaled by the rod lengths (Sec. 5).

None of the ingredients is new in isolation. Daniel Bernoulli's 1732 memoir on the small oscillations of bodies attached to a flexible thread and of the vertically hanging chain already contains both the discrete and the continuous problem,[8] the beaded string and its oscillation properties are treated in the classic monograph of Gantmacher and Krein,[9] the continuous hanging chain is a textbook problem,[10] hanging chains of discrete links have been analyzed in this journal, by McCreesh *et al.* in terms of the displacements of the pivots and by Levinson in terms of the link angles for a general class of links,[11, 12] the continuum equation has been derived as the limit of the discrete one,[13, 14] a tridiagonal system for the tensions of a discrete chain appears in Preston's analysis of whips and chains,[15] and computational multibody dynamics achieves cost proportional to n for any serial chain through recursive or sparse algorithms.[16–19] Our purpose is to connect these results in a single, elementary thread, written in the angle coordinates in which students first meet the problem, with explicit formulas that can be checked by hand for $n = 2$ and coded in a few lines. We have not found the identification of the zero-angle tension matrix with the inverse mass matrix (Corollary 1) stated explicitly. Suggested problems are collected in Sec. 6.

2 Equations of motion in angle coordinates

Consider point masses $m_1, \dots, m_n$ joined by massless rigid rods of lengths $l_1, \dots, l_n$, with the first rod hinged at a fixed pivot (Fig. 1). With θ_k the angle of rod k from the downward vertical, mass i is at $x_i = \sum_{k \le i} l_k \sin\theta_k$, $y_i = -\sum_{k \le i} l_k \cos\theta_k$, and its squared speed is $v_i^2 = \sum_{k,r \le i} l_k l_r \dot\theta_k \dot\theta_r \cos(\theta_k - \theta_r)$. Mass i contributes to the pair (k, r) only if both links lie above it, that is, if $i \ge \max(k, r)$. Exchanging the order of summation gives the following.

Proposition 1. *With the suffix masses $S_j = \sum_{i=j}^{n} m_i$ (the mass hanging below the top of rod j),*

$$T = \frac{1}{2} \sum_{k,r=1}^{n} S_{\max(k,r)}\, l_k l_r \cos(\theta_k - \theta_r)\, \dot\theta_k \dot\theta_r, \qquad U = -g \sum_{k=1}^{n} S_k\, l_k \cos\theta_k. \tag{1}$$

The coefficient $S_{\max(k,r)}$, the mass hanging below both links, organizes every term of the Lagrange equations: the mass matrix $M_{kr}(\boldsymbol\theta) = S_{\max(k,r)} l_k l_r \cos(\theta_k - \theta_r)$, the velocity-squared terms obtained by differentiating the cosines, and the gravitational torques $S_k g l_k \sin\theta_k$. This regrouping is implicit in earlier derivations.[4, 6]

For small oscillations about the downward equilibrium, $\mathbf{M}\ddot{\boldsymbol\theta} + \mathbf{K}\boldsymbol\theta = \mathbf{0}$ with constant matrices

$$M_{kr} = S_{\max(k,r)}\, l_k l_r, \qquad K_{kr} = g\, S_k\, l_k\, \delta_{kr}. \tag{2}$$

Gravity does not couple different angles, but $\mathbf{M}$ is completely full; for $n = 3$ and unit masses and lengths,

$$\mathbf{M} = \begin{pmatrix} 3 & 2 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

The key step is to notice that $S_{\max(k,r)} = \sum_i m_i\, [i \ge k][i \ge r]$. With $\mathbf{E}$ the $n \times n$ lower-triangular matrix of ones ($E_{ik} = 1$ for $i \ge k$), $\mathbf{D} = \mathrm{diag}(m_i)$ and $\mathbf{L} = \mathrm{diag}(l_k)$, this reads

$$\mathbf{M} = \mathbf{L}\,\mathbf{E}^{\mathsf T}\mathbf{D}\,\mathbf{E}\,\mathbf{L}. \tag{3}$$

Equation (3) immediately shows that $\mathbf{M}$ is symmetric and positive definite, since $\mathbf{D}$ is positive definite and $\mathbf{EL}$ is invertible.

3 A beaded string in disguise

Equation (3) suggests the change of variables

$$\mathbf{x} = \mathbf{EL}\boldsymbol\theta, \qquad x_i = \sum_{k \le i} l_k \theta_k, \tag{4}$$

which is simply the linearized horizontal displacement of mass i. The same kind of coordinates, the displacements of the pivots, were used by McCreesh *et al.* for chains of uniform rods, for which they also make the matrices banded.[11] In these variables the kinetic energy is diagonal, $T = \frac{1}{2}\dot{\mathbf{x}}^{\mathsf T}\mathbf{D}\dot{\mathbf{x}} = \frac{1}{2}\sum_i m_i \dot x_i^2$. Since $\theta_j = (x_j - x_{j-1})/l_j$ with $x_0 = 0$, the potential energy becomes

$$U = \frac{1}{2} \sum_{j=1}^{n} \frac{g S_j}{l_j}\, (x_j - x_{j-1})^2. \tag{5}$$

that is, $U = \frac{1}{2}\mathbf{x}^{\mathsf T}\mathbf{K}_x\mathbf{x}$ with a tridiagonal stiffness matrix $\mathbf{K}_x$ whose diagonal entries are $gS_j/l_j + gS_{j+1}/l_{j+1}$ (with $S_{n+1} := 0$) and whose off-diagonal entries are $-gS_{j+1}/l_{j+1}$. This is the energy of a massless string loaded with beads m_i, whose segment j has length l_j and tension $\tau_j = gS_j$, the weight hanging below it: precisely the discrete problem of Bernoulli's memoir.[8] The equations of motion,

$$m_i \ddot x_i = \frac{\tau_{i+1}}{l_{i+1}}(x_{i+1} - x_i) - \frac{\tau_i}{l_i}(x_i - x_{i-1}), \qquad \tau_{n+1} := 0, \tag{6}$$

are exactly what Newton's law gives for the horizontal components of the rod tensions at small angles. Each mass is coupled only to its two neighbors.

The tridiagonal structure survives in the original angle coordinates, in a less obvious form. Inverting Eq. (3) and using the fact that $\mathbf{E}^{-1}$ takes backward differences (ones on the diagonal and -1 just below it), we obtain the following.

Theorem 1 (Tridiagonal inverse mass matrix)**.** *With $1/m_0 := 0$, $\mathbf{M}^{-1}$ is symmetric and tridiagonal, with*

$$(\mathbf{M}^{-1})_{kk} = \frac{1}{l_k^2}\Big(\frac{1}{m_k} + \frac{1}{m_{k-1}}\Big), \qquad (\mathbf{M}^{-1})_{k,k-1} = (\mathbf{M}^{-1})_{k-1,k} = -\frac{1}{m_{k-1} l_{k-1} l_k}. \tag{7}$$

Consequently the dynamical matrix $\mathbf{Q} = \mathbf{M}^{-1}\mathbf{K}$, which governs $\ddot{\boldsymbol{\theta}} = -\mathbf{Q}\boldsymbol{\theta}$, is tridiagonal too:

$$Q_{kk} = \frac{gS_k}{l_k}\Big(\frac{1}{m_k} + \frac{1}{m_{k-1}}\Big), \qquad Q_{k,k-1} = -\frac{gS_{k-1}}{m_{k-1}\, l_k}, \qquad Q_{k-1,k} = -\frac{gS_k}{m_{k-1}\, l_{k-1}}. \tag{8}$$

Proof. $\mathbf{M}^{-1} = \mathbf{L}^{-1}\mathbf{E}^{-1}\mathbf{D}^{-1}\mathbf{E}^{-\mathsf{T}}\mathbf{L}^{-1}$, and the entries of $\mathbf{E}^{-1}\mathbf{D}^{-1}\mathbf{E}^{-\mathsf{T}}$ vanish unless $|k-r| \le 1$; they equal $1/m_k + 1/m_{k-1}$ on the diagonal and $-1/m_{k-1}$ next to it. Multiplying on the right by the diagonal matrix $\mathbf{K}$ gives $\mathbf{Q}$. □

For the uniform double pendulum ($m_k = l_k = 1$), Eq. (8) gives $\mathbf{Q} = g\begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix}$, whose eigenvalues reproduce the textbook frequencies $\omega^2 = (2 \pm \sqrt{2})\, g$. The matrix $\mathbf{E}^{\mathsf{T}}\mathbf{D}\mathbf{E}$ is an example of what Gantmacher and Krein call a single-pair matrix, and the tridiagonality of the inverses of such matrices is classical.[9, 20] Theorem 1 has a practical side, too: the linear equations can be set up and advanced in time with work proportional to n, without ever forming or factorizing the dense matrix $\mathbf{M}$. Even the kinetic energy needs only a running sum, $\frac{1}{2}\dot{\boldsymbol{\theta}}^{\mathsf{T}}\mathbf{M}\dot{\boldsymbol{\theta}} = \frac{1}{2}\sum_i m_i(\sum_{k\le i} l_k\dot{\theta}_k)^2$.

4 Normal modes and the hanging chain

Because $\mathbf{K}$ is diagonal and positive, $\mathbf{Q}$ is similar to the symmetric matrix $\mathbf{J} = \mathbf{K}^{1/2}\mathbf{M}^{-1}\mathbf{K}^{1/2}$, whose diagonal equals that of $\mathbf{Q}$ and whose off-diagonal entries are

$$J_{k,k-1} = J_{k-1,k} = -\frac{g\sqrt{S_{k-1}S_k}}{m_{k-1}\sqrt{l_{k-1}l_k}} < 0. \tag{9}$$

A symmetric tridiagonal matrix with non-zero off-diagonal entries is called a Jacobi matrix, and its properties are classical.[9] Two of them matter here. The first is easily derived in class: the eigenvalues are simple, because the rows of $\mathbf{J}\mathbf{w} = \lambda\mathbf{w}$ form a three-term recurrence that fixes $\mathbf{w}$ once w_1 is chosen, so every eigenspace is one-dimensional. Hence

$$0 < \omega_1 < \omega_2 < \cdots < \omega_n$$

for every choice of masses and lengths; the multiple pendulum never has degenerate modes. The node structure of the modes requires a little more care, because this paper uses two representations, angles and horizontal displacements, and a sign-change property of one does not imply that of the other: the displacements are cumulative sums of $l_k\theta_k$, and cumulative sums can change the number of sign changes [for example, $(1, -0.2, 1)$ has two, its cumulative sums $(1, 0.8, 1.8)$ none]. We therefore state the classical result in the form needed and apply it to each representation separately.

Lemma 1 (Sign changes of Jacobi eigenvectors)**.** *Let $\mathbf{A}$ be a real symmetric tridiagonal $n \times n$ matrix whose off-diagonal entries $-b_j$ ($j = 1, \dots, n-1$) are all negative, and let $\mathbf{w}$ be an eigenvector for its k-th smallest eigenvalue λ_k. Then $w_1 \neq 0$ and $w_n \neq 0$, an interior component can vanish only between two components of opposite sign, and $\mathbf{w}$ has exactly $k-1$ sign changes, whether vanishing components are skipped or not.*

Proof. Let $p_j(\lambda) = \det(\lambda\mathbf{I} - \mathbf{A}_j)$, with $\mathbf{A}_j$ the leading $j \times j$ block and $p_0 = 1$. Substituting into the rows of $(\mathbf{A} - \lambda\mathbf{I})\mathbf{w} = \mathbf{0}$ shows that $w_j = (-1)^{j-1} p_{j-1}(\lambda)/(b_1 \cdots b_{j-1})$ is a solution, unique up to normalization. Since all $b_j > 0$, $\mathbf{w}$ changes sign between j and $j+1$ exactly where p_{j-1} and p_j have the *same* sign. By Sturm's theorem, the number of such sign agreements in $(p_0, \ldots, p_{n-1})$ at λ (with the standard convention that a vanishing p_j counts as disagreeing with p_{j-1}, so that across it exactly one sign change of $\mathbf{w}$ is counted; see the last sentence of this proof) equals the number of eigenvalues of $\mathbf{A}_{n-1}$ below λ, and by strict interlacing this number is $k-1$ at $\lambda = \lambda_k$. Strict interlacing also gives $p_{n-1}(\lambda_k) \neq 0$, so $w_n \neq 0$, while $w_1 = 1$. If $p_j(\lambda_k) = 0$ for some interior j, the recurrence $p_{j+1} = (\lambda - a_{j+1})p_j - b_j^2 p_{j-1}$ gives $p_{j+1}p_{j-1} < 0$, so the neighbors of the vanishing component have opposite signs. □

Applied to $\mathbf{J}$, whose off-diagonal entries (9) are negative, the lemma shows that the eigenvector $\mathbf{w}$ of the k-th mode has exactly $k-1$ sign changes. The angle mode shape $\mathbf{u} = \mathbf{K}^{-1/2}\mathbf{w}$ differs from it by a positive diagonal scaling, so *the k-th angle mode has $k-1$ sign changes.* For the displacements we apply the lemma independently to the beaded string: the modes satisfy $\mathbf{K}_x\mathbf{x} = \omega^2\mathbf{D}\mathbf{x}$, and $\mathbf{D}^{-1/2}\mathbf{K}_x\mathbf{D}^{-1/2}$ is a Jacobi matrix with off-diagonal entries $-gS_{j+1}/(l_{j+1}\sqrt{m_j m_{j+1}}) < 0$ and the same eigenvalues ω_k^2. Hence *the horizontal displacements $x_1, \ldots, x_n$ of the k-th mode also have exactly $k-1$ sign changes*: the chain crosses the vertical through the pivot $k-1$ times between the pivot and the free end, the pivot itself ($x_0 = 0$) not being counted, as for a vibrating string and for the $k-1$ interior zeros of the continuum modes below.

For a uniform chain of total mass m_{tot} and length L ($m_k = m_{\mathrm{tot}}/n$, $l_k = L/n$), Eq. (6) is a finite-difference version of the hanging-chain equation $\partial_t^2 u = \partial_z(gz\,\partial_z u)$, where z is the height above the free end. Its modes are $u_k(z) = J_0(j_{0,k}\sqrt{z/L})$, with frequencies $\omega_k = \frac{1}{2} j_{0,k}\sqrt{g/L}$, where $j_{0,k}$ is the k-th zero of the Bessel function J_0.[10] Figure 2(b) shows that the discrete frequencies lie below these values and that, asymptotically for $n \gg k$, the relative error approaches $1/(4n)$; for small n and higher modes the error is smaller (at $n = 16$ it is 0.0125 for the third mode against $1/(4n) = 0.0156$, and at $n = 4$ the third-mode error is about four times smaller than $1/(4n)$). Since $\omega \propto L^{-1/2}$, the asymptotic error is that of a continuous chain lengthened by half a link, $L + l/2$ with $l = L/n$. The same picture accounts for the mode shapes if the free end of this effective continuum is placed $l/2$ below the lowest mass. Normalizing discrete and continuous displacements to one at the lowest mass, the largest difference for $n = 40$ is 0.001, 0.04, 0.12 and 0.25 for the first four modes when the continuum of length L ends at the lowest mass, but only $< 10^{-3}$, 0.001, 0.007 and 0.034 for the continuum of length $L + l/2$ [Fig. 2(a)]. This observation fits naturally with earlier work on chains of rigid links. In Levinson's angle formulation, the mass center of each link lies a fraction d of the link length below its upper pin, and the link has radius of gyration $\sqrt{k}\,l$; our point-mass chain is the case $d = 1$, $k = 0$, for which his matrices reduce, up to constant factors, to Eq. (2) for a uniform chain.[12] His tables show that the convergence to the continuum depends on d and k, and that uniform rods ($d = \frac{1}{2}$) converge rapidly, as also found by McCreesh *et al.*[11] Computing with Levinson's matrices, we find that for $d = 1$, $\frac{3}{4}$ and $\frac{1}{2}$ the relative frequency error approaches $-(d - \frac{1}{2})/(2n)$, the error of a continuum of length $L + (d - \frac{1}{2})\,l$, and that this $O(1/n)$ term does not depend on the radius of gyration (we checked $k = 0$, 0.05, 0.1 and $\frac{1}{12}$). For uniform rods ($d = \frac{1}{2}$, $k = \frac{1}{12}$) the $O(1/n)$ error therefore vanishes, leaving an $O(1/n^2)$ error, which explains their rapid convergence; for point masses at the lower ends of the links, it is $-1/(4n)$. We present these as numerical observations, without a claim of priority, and leave a heuristic explanation to the reader (Problem 4).

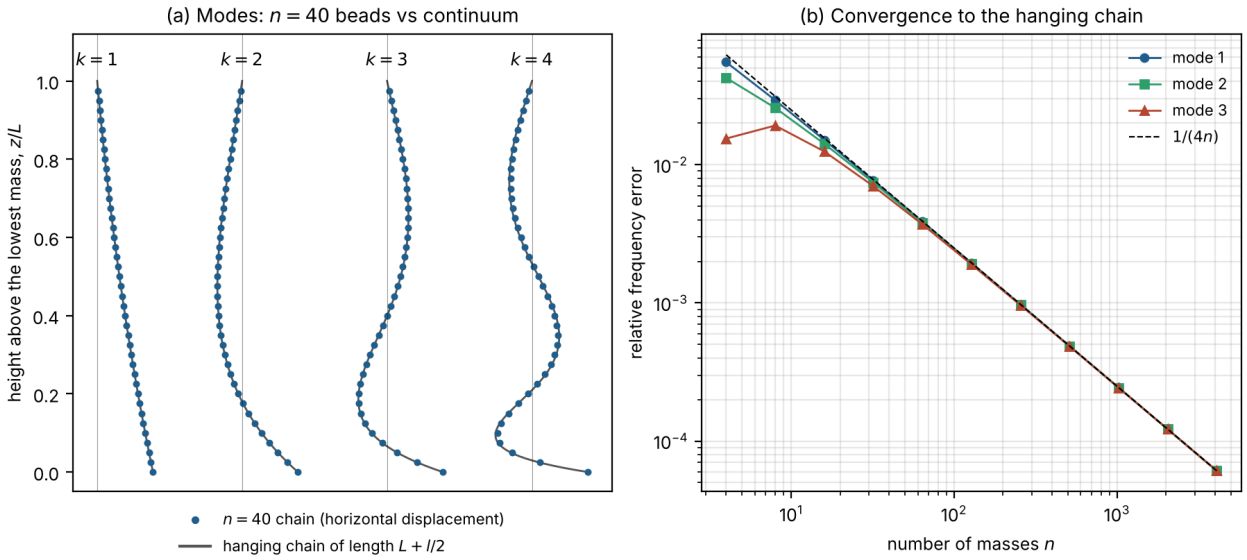


Figure 2: (a) First four normal modes of a uniform chain of $n = 40$ masses, plotted as the horizontal displacements of the masses (dots), compared with the modes of a continuous hanging chain of length $L + l/2$ whose free end lies $l/2$ below the lowest mass (lines); both are normalized to one at the lowest mass, at height $z = 0$. Mode k has $k - 1$ nodes. (b) Relative error of the three lowest frequencies with respect to the hanging chain, $\omega_k = \frac{1}{2} j_{0,k} \sqrt{g/L}$; the dashed line is $1/(4n)$.

5 Exact nonlinear dynamics: tensions

At large amplitudes the equations become $\mathbf{M}(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}} + \mathbf{c}(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}) + \mathbf{g}(\boldsymbol{\theta}) = \mathbf{0}$ with the dense, angle-dependent $\mathbf{M}(\boldsymbol{\theta})$ of Sec. 2. The beaded-string picture suggests a different route: write Newton's law for each mass, with the rod tensions τ_k as unknowns, as in Preston's treatment of chains.[15] Let $\mathbf{e}_k = (\sin\theta_k, -\cos\theta_k)$ be the unit vector along rod k, pointing from mass $k-1$ to mass k (mass 0 is the pivot), and $\mathbf{n}_k = (\cos\theta_k, \sin\theta_k)$ its normal. Then

$$m_i \ddot{\mathbf{r}}_i = -\tau_i \mathbf{e}_i + \tau_{i+1} \mathbf{e}_{i+1} - m_i g \hat{\mathbf{y}}, \qquad \ddot{\mathbf{r}}_k - \ddot{\mathbf{r}}_{k-1} = l_k \ddot{\theta}_k \, \mathbf{n}_k - l_k \dot{\theta}_k^2 \, \mathbf{e}_k, \tag{10}$$

with $\tau_{n+1} = 0$ and $\ddot{\mathbf{r}}_0 = \mathbf{0}$. Dividing the first equation by m_i, subtracting it for consecutive masses and projecting the result on $\mathbf{e}_k$ and $\mathbf{n}_k$ gives the following.

Theorem 2 (Tension formulation)**.** *The tensions solve the symmetric, positive-definite, tridiagonal system* $\mathbf{B}(\boldsymbol{\theta})\boldsymbol{\tau} = \mathbf{r}$*,*

$$B_{kk} = \frac{1}{m_k} + \frac{1}{m_{k-1}}, \qquad B_{k,k+1} = B_{k+1,k} = -\frac{\cos(\theta_{k+1} - \theta_k)}{m_k}, \qquad r_k = l_k \dot{\theta}_k^2 + g \cos\theta_1 \, \delta_{k1}, \tag{11}$$

and the angular accelerations then follow explicitly:

$$l_k \ddot{\theta}_k = \frac{\tau_{k+1} \sin(\theta_{k+1} - \theta_k)}{m_k} - \frac{\tau_{k-1} \sin(\theta_k - \theta_{k-1})}{m_{k-1}} - g \sin\theta_1 \, \delta_{k1}. \tag{12}$$

Proof. The projections use $\mathbf{e}_j \cdot \mathbf{e}_k = \cos(\theta_j - \theta_k)$, $\mathbf{e}_j \cdot \mathbf{n}_k = \sin(\theta_j - \theta_k)$, $\hat{\mathbf{y}} \cdot \mathbf{e}_1 = -\cos\theta_1$ and $\hat{\mathbf{y}} \cdot \mathbf{n}_1 = \sin\theta_1$. Positive definiteness follows from $\mathbf{B} = \mathbf{G}\boldsymbol{\mathcal{M}}^{-1}\mathbf{G}^{\mathsf{T}}$, where $\boldsymbol{\mathcal{M}} = \mathrm{diag}(m_i \mathbf{I}_2)$ and row k of $\mathbf{G}$ contains $\mathbf{e}_k^{\mathsf{T}}$ in the block of mass k and $-\mathbf{e}_k^{\mathsf{T}}$ in the block of mass $k-1$. $\mathbf{G}$ has full row rank: row n is the only row that involves mass n, so it is independent of the others; removing it, row $n-1$ is the only remaining row that involves mass $n-1$, and so on. □

The tensions are those of rigid rods and can become negative (compression) in violent motions, which a rope or chain could not sustain (Problem 6).

Corollary 1 (The linear theory as the zero-angle limit)**.** *At* $\boldsymbol{\theta} = \mathbf{0}$ *all cosines equal one, and*

$$\mathbf{B}(\mathbf{0}) = \mathbf{E}^{-1}\mathbf{D}^{-1}\mathbf{E}^{-\mathsf{T}} = \mathbf{L}\,\mathbf{M}^{-1}\mathbf{L}.$$

At rest $\boldsymbol{\tau} = g\,\mathbf{E}^{\mathsf{T}}\mathbf{D}\mathbf{E}\,\hat{\mathbf{e}}_1$*, so* $\tau_k = gS_k$*, which are the beaded-string tensions of Sec. 3. Linearizing Eq. (12) about rest gives* $\ddot{\boldsymbol{\theta}} = -\mathbf{Q}\boldsymbol{\theta}$ *with* $\mathbf{Q}$ *of Eq. (8).*

Proof. For the linearization, $\dot{\theta}_k^2$ and $\tau_k - gS_k$ are of second order (the tensions are even under $\boldsymbol{\theta} \to -\boldsymbol{\theta}$, $\dot{\boldsymbol{\theta}} \to -\dot{\boldsymbol{\theta}}$), so to first order $l_k\ddot{\theta}_k = gS_{k+1}(\theta_{k+1} - \theta_k)/m_k - gS_{k-1}(\theta_k - \theta_{k-1})/m_{k-1} - g\theta_1\delta_{k1}$. Using $S_{k+1} = S_k - m_k$ and $S_{k-1} = S_k + m_{k-1}$ reproduces each entry of Eq. (8). □

The inverse mass matrix of the small-oscillation problem is thus the tension matrix of the full nonlinear problem frozen at zero angle, rescaled by the rod lengths. The reason is simple: at rest all rods are vertical, so the tensions act vertically while the small motions are horizontal, and both are transmitted from mass to mass by the same backward-difference operator $\mathbf{E}^{-1}$, which appears in $\mathbf{B}(\mathbf{0}) = \mathbf{E}^{-1}\mathbf{D}^{-1}\mathbf{E}^{-\mathsf{T}}$ and in $\mathbf{L}\mathbf{M}^{-1}\mathbf{L}$ alike. In the angle description, the nearest-neighbor coupling of Theorem 1 is the small-amplitude shadow of the fact that each mass is held only by two rods.

Theorem 2 also gives a simple algorithm: one tridiagonal solve and one explicit formula per evaluation, about ten lines of code (Appendix). Recursive multibody algorithms reach the same cost proportional to n for general serial chains,[16–18] and constrained integrators of the SHAKE/RATTLE family exploit the same tridiagonal matrices for chains.[19, 21] The point here is transparency rather than speed records. Figure 3(a) compares a Python implementation of Eqs. (11) and (12) with forming and factorizing the dense $\mathbf{M}(\boldsymbol{\theta})$. The two agree to a relative difference below 5×10^{-12} for $n \le 3000$. For a few tens of masses or fewer the dense route is faster, because the fixed overhead of the interpreter dominates (the crossover lies between $n = 30$ and $n = 100$ on our machine), while at $n = 3000$ the tension route takes a few times $10^{-4}\,\mathrm{s}$ against $0.6\,\mathrm{s}$. As a check that connects the two halves of the paper, Fig. 3(b) shows that a uniform 50-mass chain released from its fundamental mode with amplitude ε deviates from the linear normal-mode solution by an amount that, relative to ε, scales as ε^2 (observed exponent 1.99), as expected from the cubic leading nonlinearity. The accelerations were integrated with the adaptive Runge–Kutta method DOP853.[22]

6 Suggested problems

1. Expand the kinetic energy of the double and triple pendulum and verify Proposition 1 and the factorization of Eq. (3).

2. Derive Eq. (6) directly from Newton's law, using the horizontal components of the rod tensions $\tau_j = gS_j$ at small angles, without passing through the Lagrangian.

3. Invert the 2×2 and 3×3 mass matrices of a pendulum with arbitrary masses and lengths and compare with Eq. (7). Use Eq. (8) to find the frequencies of a double pendulum with $m_1 \neq m_2$ and check the limits $m_2 \to 0$ and $m_2 \to \infty$. (In the second limit one frequency diverges and the other tends to $\sqrt{g/(l_1 + l_2)}$, that of a heavy bob on a rigid rod of length $l_1 + l_2$.)

4. Compute the lowest frequencies of uniform chains for $n = 4, \ldots, 4096$ with a tridiagonal eigensolver applied to $\mathbf{J}$ (for example, `scipy.linalg.eigh_tridiagonal`) and show that n

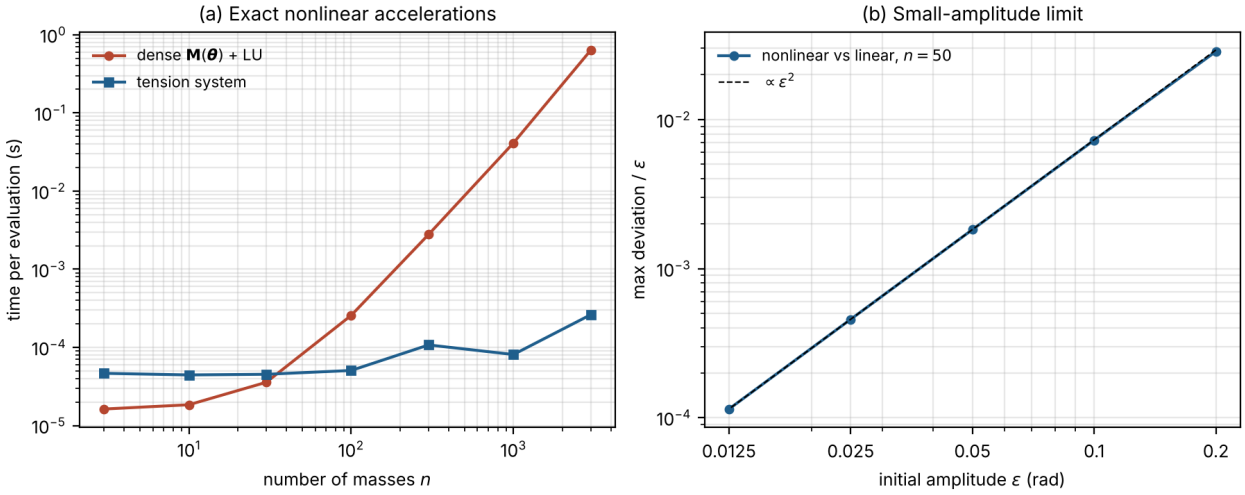


Figure 3: (a) Time for one evaluation of the exact nonlinear accelerations (single thread, Python/NumPy): dense mass matrix with LU factorization versus the tension system of Theorem 2. (b) Maximum deviation, over $0 \le t \le 5\,\mathrm{s}$, of the nonlinear motion of a uniform 50-mass chain ($m_{\mathrm{tot}} = 1\,\mathrm{kg}$, $L = 1\,\mathrm{m}$) from the linear normal-mode solution, divided by the initial amplitude ε of the fundamental mode.

times the relative error with respect to the hanging chain tends to $1/4$. Give a heuristic explanation of the effective length $L + l/2$. (Hint: in the discrete chain, the mass of each segment is concentrated at its lower end, while the tension in segment j supports everything below its upper end. Compare the tension at a given height with the continuum value $\rho g z$.) Using Levinson's matrices,[12] repeat the computation with the mass center of each link a fraction d of the way down the link, and check that the effective length becomes $L + (d - \frac{1}{2})\, l$. (Levinson's mass matrix is dense, so use a dense generalized eigensolver such as `scipy.linalg.eigh` and n up to about 500.)

5. Implement Eqs. (11) and (12) with a tridiagonal solver and integrate a triple pendulum released from the horizontal. Monitor the energy, and compare the running time with that of the dense formulation as n grows.

6. Find a motion of the triple pendulum in which some tension τ_k becomes negative. (Hint: release a uniform triple pendulum from the horizontal position.) Could a chain of beads on a string perform this motion?

7 Conclusions

The dense coupling of the multiple pendulum in angle coordinates is a property of the description, not of the physics. A single factorization, $\mathbf{M} = \mathbf{L}\mathbf{E}^{\mathsf{T}}\mathbf{D}\mathbf{E}\mathbf{L}$, shows that the small oscillations are those of a beaded string under the tensions gS_j. It gives tridiagonal formulas for $\mathbf{M}^{-1}$ and $\mathbf{M}^{-1}\mathbf{K}$ directly in the angles, and lets classical results on Jacobi matrices deliver the qualitative structure of the normal modes. For large amplitudes the rod tensions satisfy a tridiagonal system whose zero-angle limit is the inverse mass matrix rescaled by the rod lengths, so the linear and nonlinear descriptions fit together and the exact dynamics can be coded in a few lines at a cost proportional to n. Natural extensions, suitable for projects, include rigid links with distributed mass (for which

the structure becomes block tridiagonal), damping and driving, and experiments that compare measured mode frequencies of a physical chain with the hanging-chain limit.

Author declarations

Conflict of interest. The authors have no conflicts to disclose.
Data and code availability. Python code that reproduces all figures and checks every formula in this paper is openly available at `https://github.com/yonatanorion/Unified-n-Mass-Pendulum-Operator-Framework` and archived at Zenodo, doi:10.5281/zenodo.20102179.
Use of AI tools. An AI assistant (Claude, Anthropic) was used to help verify the literature and references, write and test code, run the numerical checks and draft and edit parts of the text. The authors reviewed all content and take full responsibility for the paper.

Appendix: code

The following Python functions implement Eq. (8) and Theorem 2.

```
import numpy as np
from scipy.linalg import solveh_banded

def dynamical_matrix(m, l, g=9.81):
    """Tridiagonal Q = M^-1 K of Eq. (Q): returns (diag, sub, sup)."""
    S = np.cumsum(m[::-1])[::-1] # S_k = mass below rod k
    inv_prev = np.concatenate(([0.0], 1.0 / m[:-1]))
    diag = g * S * (1.0 / m + inv_prev) / l
    sub = -g * S[:-1] / (m[:-1] * l[1:]) # Q[k, k-1]
    sup = -g * S[1:] / (m[:-1] * l[:-1]) # Q[k-1, k]
    return diag, sub, sup

def accelerations(th, om, m, l, g=9.81):
    """Exact nonlinear angular accelerations via rod tensions."""
    inv_m = 1.0 / m
    ab = np.zeros((2, m.size))
    ab[1] = inv_m + np.concatenate(([0.0], inv_m[:-1])) # B_kk
    ab[0, 1:] = -np.cos(np.diff(th)) * inv_m[:-1] # B_k,k+1
    r = l * om**2
    r[0] += g * np.cos(th[0])
    tau = solveh_banded(ab, r) if m.size > 1 else r / ab[1] # tensions
    s = np.sin(np.diff(th))
    acc = np.zeros_like(th)
    acc[:-1] += tau[1:] * s * inv_m[:-1]
    acc[1:] -= tau[:-1] * s * inv_m[:-1]
    acc[0] -= g * np.sin(th[0])
    return acc / l
```

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